```
class Area:
    def __init__(self, length, breadth):
        self.length = length
        self.breadth = breadth

    def returnArea(self):
        return self.length * self.breadth


length = float(input("Enter length: "))
breadth = float(input("Enter breadth: "))

obj = Area(length, breadth)

print("Area of Rectangle =", obj.returnArea())
```

```
Enter length: 78
Enter breadth: 98
Area of Rectangle = 7644.0
```

```
from abc import ABC, abstractmethod

class Test(ABC):

    @abstractmethod
    def m1(self):
        pass

    @abstractmethod
    def m2(self):
        pass


class MyPython(Test):

    def m1(self):
        print(" m1 implementation")
```

```
    def m2(self):
        print(" m2 implimentation")


M= MyPython()
M.m1()
M.m2()
```

```
 m1 implementation
 m2 implimentation
```

```
from abc import ABC, abstractmethod

class Test(ABC):
    def m1(self):
        pass

class MyPython(Test):
    def m1(self):
        print("m1 implementation")

    @abstractmethod
    def m2(self):
        pass

class Child(MyPython):
    def m2(self):
        print("m2 implementation")


m = Child()
m.m1()
m.m2()
```

```
m1 implementation
m2 implementation
```

```
# Inside the abstract class we can declare the constructor
# Abstract class constructor is executed but object is not created

from abc import ABC, abstractmethod
```

```
class Test(ABC):
    def __init__(self):
        print("Constructor is called...")

    @abstractmethod
    def m1(self):
        pass


class MyPython(Test):
    def m1(self):
        print("m1 Method implementation")


# m = MyPython()
# m.m1()

m = MyPython()
m.m1()
```

```
Constructor is called...
m1 Method implementation
```

```
class Test1:
    def __init__(self):
        self.a = 101
        self._b = 201
        self.__c = 301

    def show1(self):
        print("a =", self.a)
        print("b =", self._b)
        print("c =", self.__c)


class Test2(Test1):
    def show2(self):
        print("a =", self.a)
        print("b =", self._b)
        # print("c =", self.__c)
```

```
class Test3(Test2):
    def show3(self):
        print("a =", self.a)
        print("b =", self._b)
        # print("c =", self.__c)


t1 = Test1()
t1.show1()

t2 = Test2()
t2.show2()

t3 = Test3()
t3.show3()
```

```
a = 101
b = 201
c = 301
a = 101
b = 201
a = 101
b = 201
```

```
numbers = [7, 3, 9, 2, 8]

numbers.sort()

print("Second largest number is:", numbers[-2])
```

```
Second largest number is: 8
```

```
numbers = [0,1,2,6,7]

result = []

for i in numbers:
    if i != 0:
        result.append(i)
```

```
for i in numbers:
    if i == 0:
        result.append(i)

print("After moving zeros:", result)
```

```
After moving zeros: [1, 2, 6, 7, 0]
```

```
list1 = [1, 2, 3, 4, 5]
list2 = [2, 3, 5, 6, 7]
list3 = [3, 5, 8, 9]

result = set(list1) & set(list2) & set(list3)

print(result)
```

```
{3, 5}
```

```
a = [1, 1, 0, 1, 1, 1, 0, 1]

count = 0
maximum = 0

for i in a:
    if i == 1:
        count += 1
        if count > maximum:
            maximum = count
    else:
        count = 0

print("Maximum consecutive ones:", maximum)
```

```
A = [1, 2, 3, 4, 5, 6, 7, 8, 9]

A[::2] = [10, 20, 30, 40, 50]

print(A)
```

```
[10, 2, 20, 4, 30, 6, 40, 8, 50]
```

```
a=[1,2,3,4,5]


print(a[3:0:-1])
```

```
[4, 3, 2]
```

```
arr = [[1, 2, 3, 4],
       [4, 5, 6, 7],
       [8, 9, 10, 11],
       [12, 13, 14, 15]]

for i in range(0, 4):  # i = 0 to 3
    print(arr[i].pop())  # 4, 7, 11, 15
```

```
4
7
11
15
```

```
arr = [1, 2, 3, 4, 5, 6]

for i in range(1, 6):
    arr[i-1] = arr[i]

for i in range(0, 6):
    print(arr[i], end=" ")
```

```
2 3 4 5 6 6
```

```
def f(i, values=[]):
    if values is None:
        values = []

    values.append(i)
```

```
    print(values)

# Calling the function
f(1)
f(2)
f(3)
```

```
[1]
[1, 2]
[1, 2, 3]
```

```
fruit_list1 = ['Apple', 'Berry', 'Cherry', 'Papaya']

fruit_list2 = fruit_list1
fruit_list3 = fruit_list1[:]

fruit_list2[0] = 'Guava'
fruit_list3[1] = 'Kiwi'

sum = 0

for ls in (fruit_list1, fruit_list2, fruit_list3):
    if ls[0] == 'Guava':
        sum += 1

    if ls[1] == 'Kiwi':
        sum += 20

print(sum)
```

```
22
```

```
data = [[[1, 2], [3, 4]], [[5, 6], [7, 8]]]

def fun(m):
    v = m[0][0]

    for row in m:
        for element in row:
            if v < element:
```

```
                v = element

    return v

print(fun(data[0]))
```

```
4
```

```
a = {'a': 1, 'b': 2, 'c': 3}

print(a['a'], a['b'])
```

```
1 2
```

```
fruit = {}

def addone(index):
    if index in fruit:
        fruit[index] += 1
    else:
        fruit[index] = 1

addone('Mango')
addone('Apple')
addone('Mango')
addone('Banana')

print(len(fruit))
print(fruit)
```

```
3
{'Mango': 2, 'Apple': 1, 'Banana': 1}
```

```
arr = {}

arr[1] = 1

arr[1] = 2
print(arr)

arr[1] += 1
```

```
sum = 0

for k in arr:
    sum += arr[k]

print(sum)
```

```
{1: 2}
3
```

```
my_dict = {}

my_dict[1] = 1

my_dict['1'] = 2

my_dict[1.0] = 4

print(my_dict)

sum = 0

for k in my_dict:
    sum += my_dict[k]

print(sum)
```

```
{1: 4, '1': 2}
6
```

```
my_dict = {}

my_dict[(1, 2, 4)] = 8

my_dict[(4, 2, 1)] = 10

my_dict[(1, 2)] = 12

sum = 0
```

```
for k in my_dict:
    sum += my_dict[k]

print(sum)

print(my_dict)
```

```
30
{(1, 2, 4): 8, (4, 2, 1): 10, (1, 2): 12}
```

```
box = {}

jars = {}

crates = {}

box['biscuit'] = 1
box['cake'] = 3

jars['jam'] = 4

crates['box'] = box
crates['jars'] = jars

print(len(crates['box']))
```

```
2
```

```
my_dict = {'c': 97, 'a': 96, 'b': 98}

for key in sorted(my_dict):
    print(my_dict[key])
```

```
96
98
97
```

```
rec = {"Name": "Python", "Age":"20"}

r = rec.copy()

print(id(r) == id(rec))
```

```
False
```

```
rec = {"Name" : "Python", "Age":"20"}

r = rec.copy()

print(id(r) == id(rec))

print(id(r))

print(id(rec))
```

```
False
140299137592000
140299137153792
```

```
rec = {
    "Name": "python",
    "Age": "20",
    "Addr": "NJ",
    "Country": "USA"
}

id1 = id(rec)

del rec

rec = {
    "Name": "Python",
    "Age": "20",
    "Addr": "NJ",
    "Country": "USA"
}

id2 = id(rec)
```

```
print(id1 == id2)
```

```
False
```

```
rollno = [3, 5, 7, 1, 11, 4, 5, 2]

for x in rollno:
    if x == 2 or x == 4 or x == 6 or x == 8 or x == 10:
        print("Which even number is found:", x)
        # break
```

```
Which even number is found: 4
Which even number is found: 2
```

```
rollno = [5, 4, 2, 1]

for x in rollno:
    print(x)
```

```
5
4
2
1
```

```
arr = [1, 2, 4, 5]

for x in reversed(arr):
    print(x)
```

```
5
4
2
1
```

```
# Write data to text files
# 1. write()
```

```
f = open("myfile.txt", 'w')

f.write("Ashish\n")
f.write("Amar\n")
f.write("Akash\n")

print("Write operation done successfully")

f.close()
```

```
Write operation done successfully
```

```
# 2. writelines()

f = open("myfile.txt", 'w')

l = ["Neha\n", "Alisha\n", "Amar\n", "Aditya\n", "Ashish\n"]

f.writelines(l)

print("Write operation done successfully")

f.close()
```

```
Write operation done successfully
```

```
# Read data from text files

f = open("myfile.txt", 'r')

data = f.read()

print(data)

f.close()
```

```
Neha
Alisha
Amar
Aditya
```

```
Ashish
```

```
f = open("myfile.txt", "r")

l1 = f.readlines()

print(l1)

for x in l1:
    print(x)

f.close()
```

```
['Neha\n', 'Alisha\n', 'Amar\n', 'Aditya\n', 'Ashish\n']
Neha

Alisha

Amar

Aditya

Ashish
```

```
# with statement

with open("myfile.txt", "w") as f:
    f.write("Python\n")
    f.write("is\n")
    f.write("easy\n")
    f.write("language\n")
    print("is file closed ?", f.closed)

print("Now is file closed ?", f.closed)
```

```
is file closed ? False
Now is file closed ? True
```

```
#tell()
f = open("myfile.txt", 'r')

print(f.tell())
print(f.read(2))
print(f.tell())
print(f.read(4))
print(f.tell())
```

```
0
Py
2
thon
6
```

```
# We can use seek() function to move cursor (file pointer)
# specified location.

data = "Learning python is very easy from Ashish Sir"

f = open("myfile.txt", "w")
f.write(data)
f.close()

with open("myfile.txt", "r+") as f:
    fileData = f.read()
    print(fileData)

    print("Current Cursor Position is ", f.tell())

    f.seek(15)

    print("Current Cursor Position is ", f.tell())

    f.write("-JAVA-")

    f.seek(0)
    fileData = f.read()
    print(fileData)
```

```
Learning python is very easy from Ashish Sir
Current Cursor Position is  44
Current Cursor Position is  15
Learning python-JAVA-ry easy from Ashish Sir
```

```
# WAP to print number of lines, words, characters present in a file

import os, sys

fname = input("Enter file name : ")

if os.path.isfile(fname):
    print("File exist : ", fname)
    f = open(fname, 'r')
else:
    print("File is not exist : ", fname)
    sys.exit(0)

lcount = wcount = ccount = 0

for line in f:
    lcount = lcount + 1
    ccount = ccount + len(line)
    words = line.split()
    wcount = wcount + len(words)

print("Total Lines : ", lcount)
print("Total words : ", wcount)
print("Total characters : ", ccount)

f.close()
```

```
Enter file name : Myfile
File is not exist :  Myfile
An exception has occurred, use %tb to see the full traceback.

SystemExit: 0
```